Deep Learning

Assignment - 1

(b)

Submitted to,
Prof. Archa AT

Submitted by,
Kavya Chandran GR
S3 MCA
Roll No: 23

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RAMSONS

Q. Impliment XOR fn using MP neurons.

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	0

XOR fn is represented as $y = x_1\bar{x}_2 + \bar{x}_1 x_2$

$$y = x_1\bar{x}_2 + \bar{x}_1 x_2$$
$$= Z_1 + Z_2$$
$$Z_1 \cup Z_2$$

Compute the net inputs $\omega_1 = 1, \omega_2 = -1$

(0,0) $Z_{1in} = 0\times1 + 0\times-1 = 0$

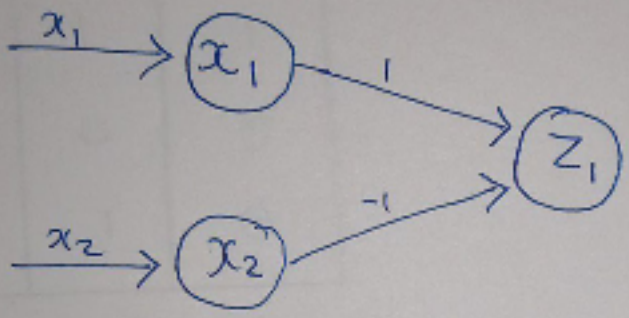
(0,1) $Z_{1in} = 0\times1 + 1\times-1 = -1$

(1,0) $Z_{1in} = 1\times1 + 0\times-1 = 1$

(1,1) $Z_{in} = 1\times1 + 1\times-1 = 0$

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On the basis of the input that is possible to get the required output Hence $w_{11} = 1$ $w_{21} = -1$ condition $\theta \geq 1$ for the Z_1 neurons



$Z_1 = x_1 \bar{x}_2$

x_1	x_2	$\bar{x}_2$	Z_1
0	0	1	0
0	1	0	0
1	0	1	1
1	1	0	0

RAMSONS

$Z_2 = \overline{x_1}\, x_2$

$\overline{x_1}$	x_2	Z_2
1	0	0
1	1	1
0	0	0
0	1	0

Compute net input Z_2

Assume $w_1 = -1$ $w_2 = 1$

(0,0) $Z_{2in} = 0 \times -1 + 0 \times 1 = 0$

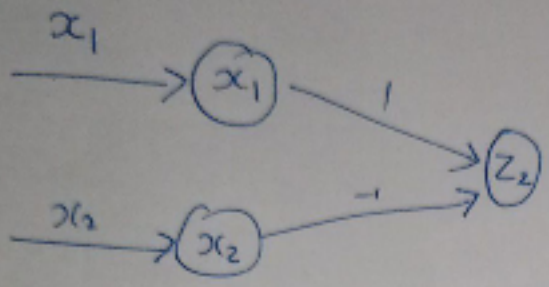
(0,1) $Z_{2in} = 0 \times -1 + 0 \times 1 = 0$

(1,0) $Z_{2in} = 1 \times -1 + 0 \times 1 = -1$

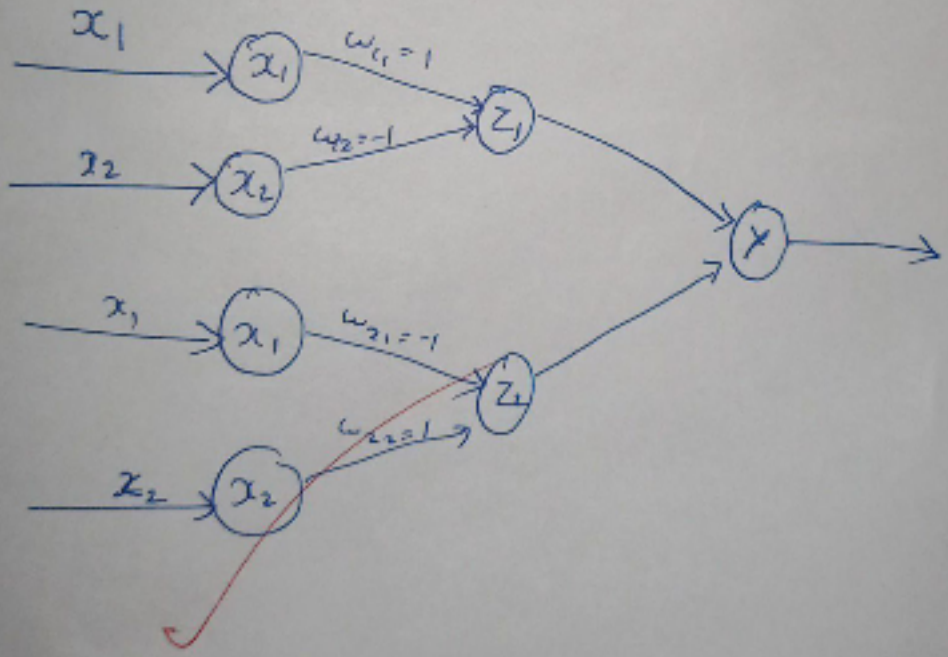
(1,1) $Z_{2in} = 1 \times -1 + 1 \times 1 = 0$

On the basic of the net input it is possible to get required output Hence $w_{21} = -1$, $w_{22} = 1$

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find neural network



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